

Eco-Scan

Problem Statement

Improper waste disposal and a lack of source segregation have triggered a severe global environmental crisis. People often want to recycle responsibly but constantly struggle to identify the correct bins for different materials. Consequently, massive amounts of recyclable plastics and e-waste end up polluting our landfills, accelerating ecological damage and wasting highly valuable natural resources.

Project Overview

Eco-Scan AI is an intelligent sustainability application designed to revolutionize everyday waste management. By leveraging deep learning and computer vision, our software instantly analyzes uploaded images to classify items into specific material categories. The system subsequently displays accurate disposal instructions and actionable environmental tips, ultimately empowering users to adopt greener habits and reduce their daily carbon footprint.

Solution Offered

The Eco-Scan consists of:

- A user-friendly Streamlit web interface for uploading images of discarded items.
- A trained neural network that accurately categorizes waste.
- Immediate onscreen guidance regarding the exact recycling bin to use.
- Educational sustainability facts that promote responsible and eco-friendly disposal habits.

Who Are The End Users?

- Environmentally conscious households wanting to separate trash accurately.
- Educational institutions teaching students about green skills.
- Municipalities optimizing local waste management programs.
- Corporate offices aiming to reduce their overall workplace environmental footprint successfully.

Technology Used To Solve The Problem

Web Interface & Deep Learning:

- **Streamlit UI:** Serves as the main user web interface.
- **TensorFlow Engine:** Drives the neural network.
- **Keras Core Framework:** Enables the building of the image classification model.

Data Processing & Application Logic:

- **Python:** The primary programming language utilized to build the application backend.